CONSTANTS

$M_E = 5.98 \cdot 10^{24}$ kg	$R_E = 6.37 \cdot 10^{6}$ m
$M_S = 1.99 \cdot 10^{30}$ kg	$R_S = 6.96 \cdot 10^{8}$ m
$M_M = 7.35 \cdot 10^{22}$ kg	$R_M = 1.74 \cdot 10^{6}$ m
$d_{E\text{-}S} = 1.50 \cdot 10^{11}$ m	$d_{E\text{-}M} = 3.84 \cdot 10^{8}$ m
$G = 6.67 \cdot 10^{-11}$ N·m^2/kg^2	$g = 9.8$ m/s^2
$k_B = 1.38 \cdot 10^{-23}$ J/K	$\sigma = 5.67 \cdot 10^{-8}$ W/m^2·K^4
$N_A = 6.02 \cdot 10^{23}$ particles/mol	$R = 8.31$ J/mol-K
$v_{\text{sound}} = 343$ m/s	$k = 1/4\pi\epsilon_0 = 8.99 \cdot 10^{9}$ N·m^2/C^2
$\rho_{\text{water}} = 1000$ kg/m^3	$\epsilon_0 = 8.85 \cdot 10^{-12}$ C^2/N·m^2
$m_p = 1.67 \cdot 10^{-27}$ kg	$\mu_0 = 4\pi \cdot 10^{-7}$ T·m/A
$m_e = 9.11 \cdot 10^{-31}$ kg	$c = 1/\sqrt{\mu_0\epsilon_0} = 3.00 \cdot 10^{8}$ m/s
$e = 1.60 \cdot 10^{-19}$ C	$h = 6.63 \cdot 10^{-34}$ J·s

Conversions

$T(°\text{C}) = T(\text{K}) - 273.15$	1 m^3 = 1000 L

1 atm = 101,300 Pa = 760 mm Hg = 14.7 psi

1 eV = $1.60 \cdot 10^{-19}$ J

Basic Formulas

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

FORCE AND ENERGY

Newton's Law

$$\sum \vec{F}_{\text{ext}} = m\vec{a} \qquad \sum \vec{\tau}_{\text{ext}} = I\vec{\alpha}$$

	Force	**Energy**
	$\vec{F} = -\vec{\nabla}U \quad \leftrightarrow$	$\Delta U = -\int \vec{F} \cdot d\vec{r}$
Gravity (on earth)	$\vec{w} = m\vec{g}$	$U_g = mgh$
Gravity (in space)	$\vec{F}_g = -\frac{Gm_1m_2}{r^2}\hat{r}$	$U_g = -\frac{Gm_1m_2}{r}$
Spring	$\vec{F}_{sp} = -k\Delta x\,\hat{x}$	$U_{sp} = \frac{1}{2}k(\Delta x)^2$
External Force	$\vec{F}_{\text{ext}}$	$W = \int \vec{F}_{\text{ext}} \cdot d\vec{r}$
Kinetic Friction	$\vec{f}_k = -\mu_k N\,\hat{v}_{\text{rel}}$	$W = \int \vec{f}_k \cdot d\vec{r}$
Drag	$\vec{D} = -(bv + cv^2)\,\hat{v}$	$W = \int \vec{D} \cdot d\vec{r}$
Buoyancy	$\vec{F}_B = -m_f\vec{g}$	$U_B = -m_f gh$
Electric	$\vec{F}_{\text{el}} = q\vec{E}$	$U_{\text{el}} = qV$
Magnetic	$\vec{F}_{\text{mag}} = q\vec{v} \times \vec{B}$	$W = 0$
Magnetic	$\vec{F}_{\text{mag-wire}} = I\vec{\ell} \times \vec{B}$	$W = 0$
Normal Force	$0 \le N \le N_{\text{max}}$	$W = 0$
Tension	$0 \le T \le T_{\text{max}}$	$W = \int \vec{T} \cdot d\vec{r}$
Static Friction	$0 \le f_s \le \mu_s N$	$W = 0$
Kinetic Energy		$K = \frac{1}{2}mv^2$
Rotational Kinetic		$K_{\text{rot}} = \frac{1}{2}I\omega^2$
Chemical Energy		U_{chem}
Thermal Energy		U_{th}

KINEMATICS

$\vec{v} = \frac{d\vec{r}}{dt}$ (slope of pos.) $\quad \vec{a} = \frac{d\vec{v}}{dt}$ (slope of vel.)

Circular Motion

$s = \theta R \quad v = \omega R \quad a = \alpha R$

$f = \frac{1}{T} \quad \omega = 2\pi f$

$a_c = v^2/R$

ROTATIONAL MOTION

$\vec{r}_{\text{cm}} = \frac{1}{m_{\text{tot}}} \sum_i m_i \vec{r}_i \qquad I = \sum_i m_i r_i^2$

$\vec{\tau} = \vec{r} \times \vec{F} \quad \tau = rF_\perp = r_\perp F = rF \sin\phi$

Moments of Inertia

$I_{\text{pt}} = I_{\text{ring}} = MR^2 \quad I_{\text{rod ab end}} = \frac{1}{3}ML^2 \quad I_{\text{rod ab mid}} = \frac{1}{12}ML^2$

$I_{\text{disk}} = \frac{1}{2}MR^2 \quad I_{\text{solid sph}} = \frac{2}{5}MR^2 \quad I_{\text{sph shell}} = \frac{2}{3}MR^2$

Parallel Axis Theorem

$I = I_{\text{cm}} + Mh^2$

Rolling

$v_{\text{cm}} = \omega R$

$a_{\text{cm}} = \alpha R$

MOMENTUM AND ENERGY

$\vec{p} = m\vec{v}$

$\vec{L} = \vec{r} \times \vec{p} \quad \vec{L} = I\vec{\omega}$

$\vec{J} = \int \vec{F}\, dt \qquad \vec{J}_{\text{rot}} = \int \vec{\tau}\, dt$

$P = \frac{dE}{dt}$ or $P = \frac{dW}{dt} = Fv$

Conservation Laws

$\vec{p}_i + \sum \vec{J}_{\text{ext}} = \vec{p}_f$

$\vec{L}_i + \sum \vec{J}_{\text{rot ext}} = \vec{L}_f$

$E_i + \sum W_{\text{ext}} + \sum Q_{\text{ext}} = E_f$

FLUIDS

$\rho = m/V$ $\quad P = F/A$

$Q = vA$ $\quad v_1 A_1 = v_2 A_2$

$P = P_0 + \rho g h$ (fluid not moving)

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$F = \eta A \frac{v}{\ell}$ $\quad Q = \frac{\pi R^4 \Delta P}{8\eta L}$

OSCILLATIONS AND WAVES

$f = \frac{1}{T}$ $\quad \omega = 2\pi f = \frac{2\pi}{T}$ $\quad k = \frac{2\pi}{\lambda}$ $\quad v = f\lambda$

General Oscillations

$x(t) = A\cos(\omega t)$ $\quad v(t) = -\omega A \sin(\omega t)$ $\quad a(t) = -\omega^2 A \cos(\omega t)$

General Waves

$D(x,t) = A\cos(kx - \omega t)$

$I = \frac{P}{A} = \frac{P_{\text{source}}}{4\pi r^2}$

Specific Examples

$\omega_{\text{mass-spring}} = \sqrt{k/m}$

$\omega_{\text{pendulum}} = \sqrt{g/L}$ $\quad \omega_{\text{phys pendulum}} = \sqrt{mgd/I}$

$v_{\text{string}} = \sqrt{T/\mu}$ $\quad v_{\text{fluid}} = \sqrt{B/\rho}$

Sound

$\beta\ (\text{in dB}) = 10\log(I/I_0)$ $\quad I_0 = 10^{-12}\ \text{W/m}^2$

$$f' = f\left(\frac{v_{\text{sound}} \pm v_{\text{obs}}}{v_{\text{sound}} \mp v_{\text{source}}}\right)$$

THERMODYNAMICS

$\rho = \frac{m}{V}$ $\quad P = \frac{F}{A}$

$E_{\text{th-ideal gas}} = \frac{c}{2}Nk_BT = \frac{c}{2}nRT$

$W = \int P\,dV$ $\quad \Delta S = \int \frac{dQ}{T}$

$PV = Nk_BT = nRT$

$PV^\gamma = \text{const}$ (adiabatic) $\quad \gamma = \frac{c+2}{c}$

$K_{\text{avg}} = \frac{3}{2}k_BT = \frac{1}{2}mv_{\text{rms}}^2$ $\quad v_{\text{rms}} = \sqrt{\frac{3k_BT}{m}}$

$\Delta L = \alpha L_0 \Delta T$ $\quad \Delta V = \beta V_0 \Delta T$

$\epsilon = \frac{W_{\text{out}}}{Q_H} = 1 - \frac{Q_C}{Q_H}$ $\quad \epsilon_{max} = 1 - \frac{T_C}{T_H}$

$COP_{\text{refr}} = COP_{\text{ac}} = \frac{Q_C}{W_{\text{in}}}$ $\quad COP_{\text{heat pump}} = \frac{Q_H}{W_{\text{in}}}$

Laws

$\Delta E_{\text{th}} = Q - W$ $\quad \Delta S_{\text{univ}} \geq 0$

Heat

$Q = mc\Delta T$ $\quad Q = mL$

$P_{\text{cond}} = \frac{Q}{\Delta t} = \frac{kA}{L}\Delta T$ $\quad P_{\text{rad}} = \frac{Q}{\Delta t} = e\sigma A T^4$

BASIC EQU. OF ELECTRICITY AND MAGNETISM

Maxwell's Equations

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \qquad \oint_S \vec{B} \cdot d\vec{A} = 0$$

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt} \qquad \oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{encl}} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

Potential and Fields

$$\vec{E} = -\vec{\nabla} V = -\frac{\partial V}{\partial x}\hat{x} - \frac{\partial V}{\partial y}\hat{y} - \frac{\partial V}{\partial z}\hat{z}$$

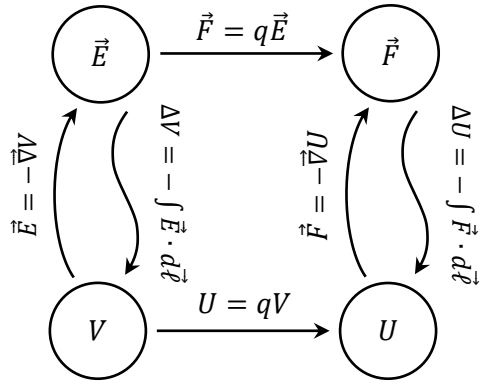
$$u_E = \frac{1}{2}\rho V = \frac{1}{2}\epsilon_0 E^2 \qquad u_B = \frac{1}{2\mu_0}B^2$$

$$\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B}$$

Other

$$\Phi_S \text{ (flux)} = \int \vec{S} \cdot d\vec{A}$$

FIELDS



Sources

$$\vec{E}_{\text{pt}} = \vec{E}_{\text{sph-outside}} = \frac{kq}{r^2}\hat{r} \qquad E_{\text{cap}} = \frac{\sigma}{\epsilon_0} \qquad E_{\text{plane}} = \frac{\sigma}{2\epsilon_0}$$

$$V_{\text{pt}} = V_{\text{sph-outside}} = \frac{kq}{r}$$

$$d\vec{B}_{\text{pt}} = \frac{\mu_0}{4\pi}\frac{Id\vec{\ell} \times \hat{r}}{r^2}$$

$$\vec{B}_{\text{wire}} = \frac{\mu_0 I}{2\pi r}\hat{\phi} \qquad \vec{B}_{\text{loop}} = \frac{\mu_0 NI}{2R}\hat{z} \qquad \vec{B}_{\text{sol}} = \mu_0 nI\,\hat{z} \qquad B_{\text{plane}} = \frac{\mu_0 K}{2}$$

Dipoles

$$\vec{p} = q\vec{d} \qquad \vec{\mu} = NI\vec{A}$$

$$\vec{\tau} = \vec{p} \times \vec{E} \qquad \vec{\tau} = \vec{\mu} \times \vec{B}$$

$$U = -\vec{p} \cdot \vec{E} \qquad U = -\vec{\mu} \cdot \vec{B}$$

Induction

$$\mathcal{E} = -\frac{d\Phi}{dt} = -L\frac{dI}{dt} \quad \text{or} \quad -M\frac{dI}{dt}$$

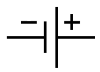
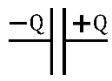
CIRCUITS

Components

$$C_{\text{parallel plate}} = \frac{\kappa\epsilon_0 A}{d} \qquad R = \frac{\rho L}{A} \qquad L = \frac{N\Phi_B}{I}$$

Kirchhoff's Laws

$$\sum I_{\text{in}} = \sum I_{\text{out}} \qquad \sum_{\text{loop}} \Delta V = 0$$

Component	$\Delta V =$	Sign	Energy
Battery	$\mathcal{E}$		$P = I\mathcal{E}$
Capacitor	Q/C		$U = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}\frac{Q^2}{C}$
Resistor	IR	ΔV drops going with I	$P = I\Delta V = I^2R = \frac{(\Delta V)^2}{R}$
Inductor	$L\frac{dI}{dt}$	ΔV drops going with I	$U = \frac{1}{2}LI^2$

AC Circuits

$$V(t) = V_0\cos(\omega t + \phi) \qquad I(t) = I_0\cos(\omega t)$$

$$V_{\text{rms}} = V_0/\sqrt{2} \qquad I_{\text{rms}} = I_0/\sqrt{2}$$

$$X_C = 1/\omega C \qquad X_L = \omega L \qquad Z = \sqrt{R^2 + (X_L - X_C)^2} \qquad \tan\phi = \frac{X_L - X_C}{R}$$

$$V_{\text{rms}} = I_{\text{rms}}Z \qquad P_{\text{avg}} = I_{\text{rms}}^2 R$$

Combination Rules

Capacitors	**Add**	**Const**
Series	$1/C$	Q
Parallel	C	V
Resistors	**Add**	**Const**
Series	R	I
Parallel	$1/R$	V
Inductors	**Add**	**Const**
Series	L	dI/dt
Parallel	$1/L$	V

Time-dependent Circuits

RC Circuits		$\boldsymbol{\tau = RC}$
Charging	$Q = \mathcal{E}C\left(1 - e^{-t/\tau}\right)$	$I = I_0 e^{-t/\tau}$
Discharging	$Q = Q_0 e^{-t/\tau}$	
LR Circuits		$\boldsymbol{\tau = L/R}$
Ramping up		$I = \frac{\mathcal{E}}{R}\left(1 - e^{-t/\tau}\right)$
Ramping down		$I = I_0 e^{-t/\tau}$
LC Circuits		$\boldsymbol{\omega_0 = 1/\sqrt{LC}}$
Oscillating	$Q = Q_0\cos\omega_0 t$	$I = Q_0\omega_0\sin\omega_0 t$

WAVE OPTICS

Electromagnetic Waves

$E(z,t) = E_0 \cos(kz - \omega t)$

$B(z,t) = B_0 \cos(kz - \omega t)$

$v = 1/\sqrt{\mu\epsilon} = c/n \qquad E_0 = cB_0$

$I = \frac{P}{A} = \frac{P_{\text{source}}}{4\pi r^2} = S_{\text{rms}} = \frac{E_{\text{rms}} B_{\text{rms}}}{\mu_0} = \frac{E_0 B_0}{2\mu_0}$

Interference

constructive: extra distance $= m\frac{\lambda}{n}$

destructive: extra distance $= \left(m + \frac{1}{2}\right)\frac{\lambda}{n}$

double slit: extra distance $= d \sin\theta_m \approx d\theta_m \quad \theta_m \approx \tan\theta_m = \frac{y_m}{L}$

thin film: extra distance $= 2t$ const/dest reverse for 1 flip

Diffraction

destructive: $a \sin\theta_p = p\frac{\lambda}{n}$ except $p = 0$

$\Delta\theta = \frac{1.22\lambda}{D}$

GEOMETRIC OPTICS

$n_1 \sin\theta_1 = n_2 \sin\theta_2 \qquad \theta_c = \sin^{-1}(n_2/n_1)$

$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \qquad m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$

$f = \frac{r}{2}$

Quantity	Positive when	Negative when
d_o	Object in front	Object behind
d_i	Image where light is (real)	Image not where light is (virtual)
f	Converging lens, concave mirror	Diverging lens, convex mirror
m	Upright	Inverted